

A. FUNCTIONAL PERFORMANCE REQUIREMENTS – GRID INTEGRATION

A.01. INTRODUCTION

This schedule specifies Technical Limits based on preliminary modelling and assessment of the Facility and Network.

In connection with its award, the Project Company will provide CEB with its proposed Technical Limits together with supporting technical information provided by the Construction contractor and equipment supplier(s) as required by the Project Company in accordance with the Specifications (including power systems model in PSS®E and PSCAD format). Any deviations to the Technical Limits herein shall be identified, and demonstration provided that these do not adversely impact the stability, reliability and power quality of the Network.

CEB may request additional data to determine the appropriateness of the Technical Limits.

CEB's feedback on the acceptability of the Project Company's proposed Technical Limits shall not be unreasonably withheld or delayed and shall be deemed to be given unless CEB informs the Project Company in writing of any disagreement (giving reasons) within sixty (60) days of receipt by it of the information provided by the Company pursuant to this Schedule.

During the Supply Period, configurable settings for Technical Limits must be reconfigured at the request of CEB to maintain stability following changes in the network. Any such changes shall be agreed with CEB.

The works shall include but not be limited to the following:

- (a) Providing technical limits of the BESS system
- (b) Grid integration and control performance complied with the specification as per this document
- (c) Providing necessary PSS®E and PSCAD models with site specific settings including Releasable User Guides (RUG)
- (d) Power Plant Controller (PPC) controlling point shall be at 33 kV POC (CT & VT point related to this point will be provided by CEB).
- (e) Performance monitoring point shall be at 33 kV (POC)

POC – Point of Connection -33 kV

A.02. DEFINITIONS, STANDARDS AND OPERATING CONDITIONS

The BESS and associated equipment shall withstand environmental conditions at the Grid substation locations specified by CEB, including wind-blown dust, rain, and

other contaminants. The buildings provided by the Project Company for the equipment shall be environment controlled and free of contaminants.

A.02.01. Definitions

Unless explicitly stated in this document, or the Contract, definitions are as per the Standards.

A.02.02. Standards

All equipment shall be installed and designed in accordance with statutory requirements. Where explicit local regulations are not applicable, all equipment and works supplied shall conform to the latest editions of the relevant international standards. The following order of precedence shall apply¹:

- (a) Grid Connection Code approved by Public Utilities Commission in Sri Lanka dated 22 July 2024
- (b) Relevant International Electrotechnical Commission (IEC) Standards
- (c) Key standards applicable to the Grid Integration are specified where relevant in the Bidding Documents.

All referenced standards must be complied with unless exception is granted by EMPLOYER. If a Functional Requirement exceeds a standard listed, then that Functional Requirement shall be adhered to.

Standard(s) referred to shall mean the current Edition / Revision together with Amendments issued.

A.02.03. Site and Environmental Data

Refer Clause 1.1.3 of Volume 4 for Site Environmental Conditions.

A.02.04. Temperature and Humidity

The BESS and associated equipment shall be rated for continuous operation and energisation at temperatures ranging from 10 °C to 40 °C. The SVS shall operate in a curtailed mode if the temperature falls outside of the operating range. The Project Company shall specify such degraded ratings in the proposal.

The design relative humidity shall be 95%, measured over a period of 24 hours.

¹ If similar standards exist in different suites of standards, the lower priority standard need not be applied. Alternatively, the Project Company may seek EMPLOYER's approval of application of the lower priority standard without application of the higher priority standard. Such approval shall not unreasonably be withheld.

A.03. CONDITIONS FOR CONNECTION OF BESS

A.03.01. Outline of Requirements

- (a) This schedule sets out details of additional requirements and conditions that the Project Company must satisfy as a condition of connection of a generating system or integrated resource system to the power system as specified in the Grid Connection Code. The Project Company shall require adhering updated requirements indicated in this schedule which may be slightly deviated from the Grid Connection Code requirements when it specifically indicated in this schedule.
- (b) This schedule also sets out the requirements and conditions which subject to clause 3 of the Grid Connection Code,
- (c) To co-operate with the CEB on technical matters when making assessment; and to provide information to the CEB.
- (d) The equipment associated with each BESS and associated auxiliary equipment must be designed to withstand without damage the range of operating conditions which may arise consistent with the system standards.
- (e) The Project Company must comply with the specification given in this schedule, Grid Connection Code, and any attached terms or conditions of agreement agreed with the CEB in accordance with a relevant provision of Grid Connection Code (in line with 3.1 (a)) before hand over the unit to CEB. Clauses in this specification (Technical Requirement) takes precedence over the relevant Grid Connection Code clauses. If any discrepancy, the Project Company must clarify it with CEB.
- (f) The Project Company must agree with performance criteria with CEB according to the given specification and Grid Connection Code requirement. Once agreed it must be recorded together as the agreed performance standard for evaluation purpose.

A.03.02. Application of settings

- (a) The Project Company must only apply settings to a control system or a protection system that are necessary to comply with performance requirements of this schedule. All settings (except internal proprietary settings) must be approved in writing by the CEB. The Project Company or equipment manufacturer must not change its setting without prior approval from CEB even though they may have ability to do so.

- (b) If the Project Company seeks approval from the CEB to apply or change a setting, then approval must not be withheld unless the CEB reasonably determines that the changed setting would cause the BESS unit to not comply with the relevant performance standard or cause an inter-regional power transfer capability to be reduced.
- (c) If the CEB reasonably determines that a setting of the BESS unit's control system or protection system needs to change to comply with the relevant performance standard or to maintain or restore an inter-regional or intra-regional power transfer capability, the CEB must consult with the Project Company to change settings. CEB shall request in writing that a setting be applied in accordance with the determination and the Project Company must be able to change those settings
- (d) The Project Company who receives such a request must arrange for the notified setting to be applied as requested and for a test to be conducted as requested. After the test, the Project Company must, on request, provide CEB with a report of a requested test, including evidence of its success or failure. Such a report of a test is confidential information.
- (e) As mentioned, in A.03.02. (a)-(d) the Project Company must not change a setting requested by the CEB without prior written agreement. If the CEB requires the Project Company to change a setting within 2 months of a previous request (after commissioning), the CEB must pay the Project Company a reasonable cost of changing the setting and conducting the tests as requested.
- (f) There are at least three levels of user access control for CEB staff shall be provided. Finalizing user access control levels shall discuss during the implementation phase.

A.03.03. Technical matters to be coordinated

- (a) The Project Company and the CEB must use all reasonable endeavours to agree upon relevant technical matters in connection of the BESS to the CEB network including:
 - i. design at the connection point
 - ii. physical layout of BESS and auxiliary device locations
 - iii. primary protection and backup protection
 - iv. control characteristics as per the specification
 - v. communications facilities as per the specification
 - vi. insulation co-ordination and lightning protection, if required
 - vii. fault levels and fault clearance
 - viii. switching and isolation facilities

- ix. interlocking and synchronizing arrangements
 - x. metering installations
- (b) The Project Company must ensure that in designing a generating system's or integrated resource system's electrical plant, including any substation for the connection of the generating system or integrated resource system to the network, to operate at the same nominal voltage as at the connection point:
- i. the plant complies with the relevant standards mentioned in this schedule and Grid Connection Code unless a provision of the Grid Connection Code allows or requires otherwise
 - ii. the earthing of the plant complies with the IEEE 80: Guide for Safety in AC Substation Grounding
 - iii. the plant is capable of withstanding, without damage the voltage impulse levels specified in this specification (Grid Connection Code Appendix A)
 - iv. the insulation levels of the plant are coordinated with the insulation levels of the network to which the BESS is connected
 - v. safety provisions in respect of the plant comply with requirements applicable to the participating jurisdiction in which the BESS is located, as notified by the CEB.
- (c) If no relevant Standard exists for the purposes of paragraph (b), the Project Company must agree with the CEB to comply with another relevant standard.

A.03.04. Provision of information

- (a) The Project Company or person who is negotiating a connection agreement with a CEB must promptly on request by CEB all data in relation to that in the Grid Connection Code or this schedule.
- (b) Three months before commissioning of BESS; or
- (c) 5 business days before commissioning of a BESS alteration that is repairing plant after a plant failure, if plant performance after the alteration will differ from performance prior to the plant failure,
- (d) Must provide:
- i. to CEB
 - i. information about the protections systems of the BESS
 - ii. information about the control systems of the BESS including:
 - i. a set of functional block diagrams, including all functions between feedback signals and generating system output or integrated resource system output or consumption
 - ii. the parameters of each functional block, including all settings, gains, time constants, delays, deadbands and limits
 - iii. the characteristics of non-linear elements

- iv. encrypted models in a form suitable for the software simulation products nominated by CEB and the Grid Connection Code
- (e) The Project Company shall provide the service to convert model to another version of PSS®E and PSCAD™ if the provided models are not compatible with the new version structure until the guaranteed lifetime of the unit. CEB shall provide a reasonable cost to the Project Company to get the new updated or complied models with a reasonable timeframe which need to be established during tender negotiations.
- (f) The information provided under paragraph (d) must contain sufficient detail for CEB to perform power system simulation studies in accordance with the requirements and circumstances specified in the Grid Connection Code.
- (g) The information provided under paragraph (d) must:
 - i. encompass all control systems that respond to voltage or frequency disturbances on the power system, and which are either integral to the production units or otherwise part of the BESS, including those applying to reactive power equipment that forms part of the generating system or integrated resource system; and
 - ii. conform with the applicable models developed in accordance with the DATA AND MODEL REQUIREMENTS TO BE COMPLIED BY INVERTER BASED RENEWABLE ENERGY PLANTS (IBRE) PRIOR TO BE INTERCONNECTED TO CEB NETWORK (Refer Annex A-I) , or an alternative model agreed with CEB to be necessary to adequately represent the relevant plant to carry out load flow and dynamic simulations and (where applicable) specialised power system studies.
- (h) The Project Company must provide to CEB information that updates the information provided under paragraph (d) and must provide to the CEB information that updates the information provided under subparagraph (b):
 - i. within 3 months after commissioning tests or other tests undertaken in accordance with clause A.05.24. (c) are completed.
 - ii. when the Generator or Integrated Resource Provider becomes aware that the information is incomplete, inaccurate or out of date; or
- (i) On request by CEB, where CEB considers that the information is incomplete, inaccurate or out of date.
 - i. The Project Company is only required to provide new information to the extent that it is different to the information previously provided under clause A.05.23.
- (j) For the purposes of BESS design, the technical information that a CEB must, if requested, provide to the Project Company in respect of a proposed connection for a generating system or integrated resource system includes:

- i. the highest expected single phase and three phase fault levels at the connection point with the generating system or integrated resource system not connected
- i. the clearing times of the existing protection systems that would clear a fault at the location at which the new connection would be connected into the existing transmission system or distribution system
- ii. the expected limits of voltage fluctuation, harmonic voltage distortion and voltage unbalance at the connection point with the generating system or integrated resource system not connected
- iii. technical information relevant to the connection point with the BESS not synchronised including equivalent source impedance information, sufficient to estimate fault levels, voltage fluctuations, harmonic voltage distortion (for harmonics relevant to the BESS) and voltage unbalance
- iv. information relating to the performance of the national grid that is reasonably necessary for the Project Company to prepare an application to connect, including:
 - a. a model of the power system, including relevant considered projects and the range of expected operating conditions, sufficient to carry out load flow and dynamic simulations; and
 - b. information on inter-regional and intra-regional power transfer capabilities and relevant plant ratings; and
- v. the CEB expected three phase fault level at the connection point for the BESS following the connection of the CEB.

(k) All information provided under this clause A.05.23. and A.05.24. must be treated as confidential information.

A.04. POWER SYSTEM CHARACTERISTICS

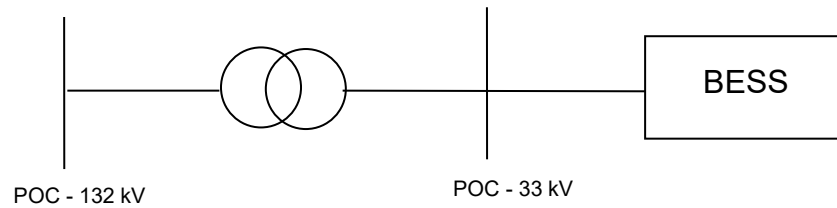


Table A1 – Necessary Power System Characteristics

No.	Type of Information	Value	Units
4	Nominal ac system voltage, line to line at POC	33	kV
5	Maximum continuous ac system voltage, line to line at POC	36	kV
6	Minimum continuous ac system voltage, line to line at POC	30	kV
7	Continuous negative-sequence voltage component (used for performance calculation)	<1 Measured as 10-minute value	%
8	Nominal ac system frequency	50	Hz
9	Maximum continuous ac system frequency (steady state)	50.5	Hz
10	Minimum continuous ac system frequency (steady state)	49.5	Hz
11	Maximum short-term ac system frequency	53	Hz
12	Minimum short-term ac system frequency	47	Hz
13	Maximum rate of change of frequency (df/dt), 500 msec averaging	2.5	Hz/s
17	Basic insulation level at POC	170	kV peak
18	Switching impulse level at POC	-	kV peak
19	Power frequency withstand voltage at POC	70	kV
21	Maximum three-phase fault current at POC (for equipment design)	25 (at 37 kV level)	kA for <u>1</u> s
22	Maximum three-phase fault current at POC	CEB will provide during the tendering process	kA
23	Minimum three-phase fault current POC		kA
26	Maximum single-phase fault current	Not Available	kA
27	Existing single-phase fault current	Not Available	kA
28	Minimum single-phase fault current	Not Available	kA
29	Background harmonic voltage (or current) spectrum (for BESS components rating)	CEB will provide during the tendering process	%
32	Normal fault clearing time at POC	250	ms
33	Breaker failure fault clearing time	550	ms

A.05. TECHNICAL REQUIREMENTS

The offered BESS shall have grid-forming control capabilities as an option, providing short-circuit capacity to the system. During steady-state and transient processes, the grid-forming converter shall instantly share unbalanced grid power, respond to voltage changes, and inject reactive power.

A.05.01. Grid Connection Code

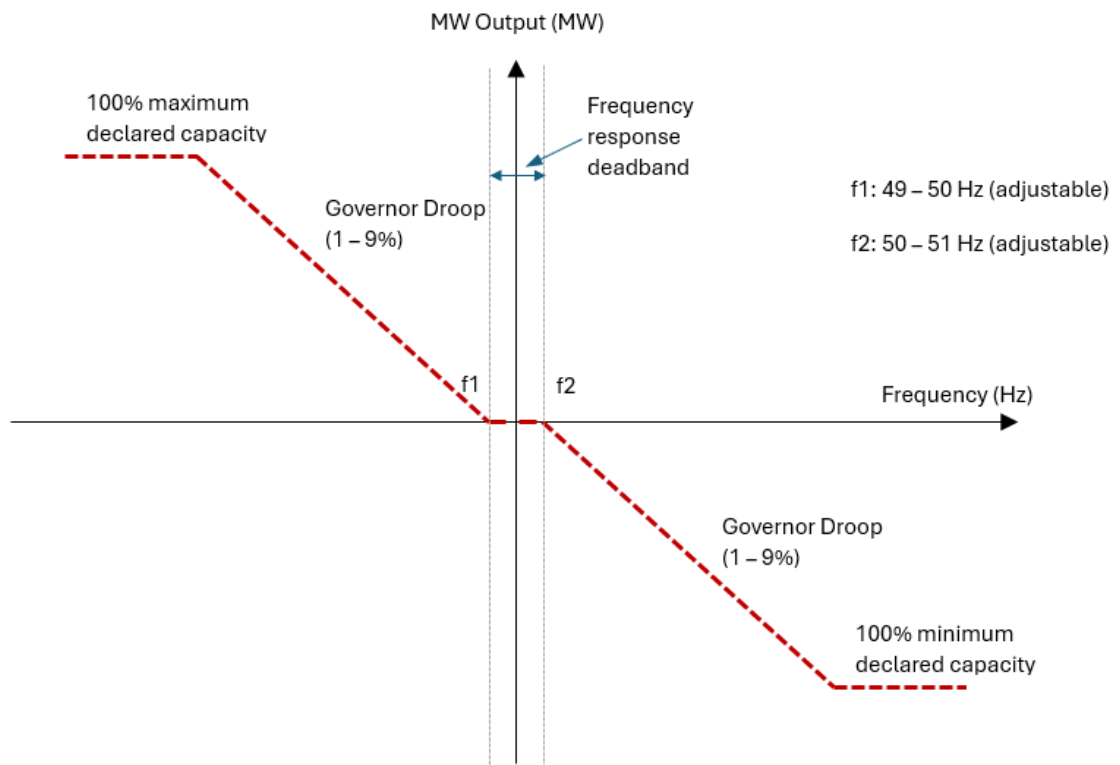
- (a) The connection code applicable for this project is Grid Connection Code approved by Public Utilities Commission in Sri Lanka dated 22 July 2024. However, some clause specifically mentioned in this Technical Requirements may differ from the Grid Connection Code and shall take precedence over the Grid Connection Code requirements. The Project Company shall clarify with CEB if there are any concerns on interpretations of the clauses.

A.05.02. Frequency variation capability

- (a) BESS shall be capable of delivering the declared active and reactive power outputs within the system Frequency variations, specified in this specification. Additionally, the PCS AC-side current of the BESS shall be capable of operating continuously at 110% of the rated current, sustain 120% of the rated current for ≥ 2 minutes, and support short-term overloads of 150% rated current for 10 seconds and 180% rated current for 100 ms.
- (b) Active and reactive power capability (± 10 MW/ ± 3 Mvar) of the BESS must be available for frequency range between 47 Hz to 52 Hz for at least for time related clause in A.05.04. If wider frequency ranges or longer minimum times for operation are economically and technically feasible, the power park module owner shall not unreasonably withhold consent.
- (c) The Project Company shall consider real and reactive power loss transformer when calculating the BESS rating at POC.
- (d) The supplied active power shall be delivered to the network based on the nominated frequency droop settings and deadband settings.
- (e) BESS shall be protected against frequency excursion outside the frequency range specified in clause in A.05.04. and able to disconnect from the system if the extreme frequency could risk to the BESS performance and reliability.

A.05.03. Frequency support capability

- (a) The BESS must be capable of responding independently on a configurable droop curve in response to frequency deviations, without receiving external control signals from CEB.
- (b) Droop set point shall be set based on studies conducted by the CEB as part of the design of the Facility for the purpose of maintaining network stability. Droop setting shall be able to set between 1% to 9% as per the CEB discretion. If CEB did not disclose any Droop value, then it shall be set at 4%.



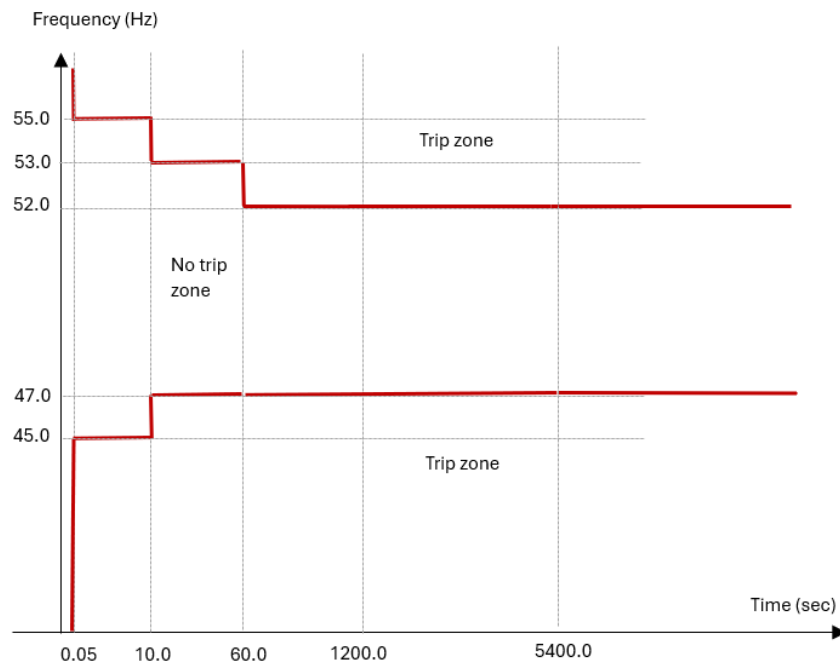
- (c) The response time for primary frequency regulation shall be <0.2 s, with active power adjustment deviation $\leq 2\%$.
- (d) No frequency regulation limit shall be set in principle, but if necessary, the limit shall be $\geq 20\%$ P_n
- (e) The adjustable frequency deadbands for f_1 shall be in the range of 49.00 Hz to 50.00 Hz, and for f_2 in the range of 50.00 Hz to 51.00 Hz. The step size (resolution) for adjusting the deadbands shall be at least 0.05 Hz.

A.05.04. Frequency withstand capability

- (a) BESS shall be capable of staying connected to the Transmission System and remain continuously operable within the System Frequency range 47 Hz to 52 Hz.
- (b) BESS must remain connected and fully operational for under frequency variations between 47 Hz to 50 Hz for duration as shown in the following table and figure. Under frequency variation window shall extend to 45 Hz to 50 Hz if equipment has the capability to do so.
- (c) BESS must remain connected and fully operational for over frequency variations between 50 Hz to 52 Hz for duration as shown in the following table and figure. Over frequency variation window shall extend to 50 Hz to 55 Hz if equipment has the capability to do so.
- (d) Frequency ride through settings shall be configurable in at least four bands between 45 Hz – 55 Hz covering above requirement. If default band range capability of equipment exceeds this requirement, the default capability must be maintained and accessible to CEB without restriction.
- (e) BESS shall change its power level up to its maximum rating according to the droop setting and frequency deviation in the system.

Frequency		Time
Hz	p.u.	
$f > 55.0$	$f > 1.10$	<i>Automatic disconnection allowed if so desired by the BESS Operator after 50 milliseconds</i>
$55.0 \geq f > 53.0$	$1.10 \geq f > 1.06$	10 seconds
$53.0 \geq f > 52.0$	$1.06 \geq f > 1.04$	1 minutes
$52 \geq f \geq 47.0$	$1.04 \geq f \geq 0.94$	Continuous operation
$47.0 > f \geq 45.0$	$0.94 > f \geq 0.9$	10 seconds
$45.0 > f$	$0.90 > f$	<i>Automatic disconnection allowed if so desired by the BESS Operator after 50 milliseconds</i>

Above frequency range specified in *Italic letters* shall consider as optional but recommended if the equipment has capability to do so.



A.05.05. Rate of change of frequency

- (a) At a minimum, the BESS shall be able to remain synchronised to the system during rate of change of system frequency of up to and including 4.0 Hz per second measured at a rolling average over 250 milliseconds and 2.5 Hz per second measured at a rolling average over 500 milliseconds.
- (b) Rate of change of frequency (RoCoF) withstand capability mentioned in the A.05.05. (a) does not mean that equipment manufacturer or BESS developer intentionally set the BESS' RoCoF protection to activate when RoCoF limit exceeds above limits in A.05.05. (a) unless violation is a risk to the device functionality or reliability. If any RoCoF protection of the BESS should be disabled unless it is essential to device stability and protection.

A.05.06. Voltage variation capability

- (a) The BESS shall be capable of delivering the declared active and reactive power outputs within the voltage variations of $\pm 10\%$ of the rated voltage at POC
- (b) The BESS shall be capable of delivering the declared rated current during under voltage condition and over voltage conditions as mentioned in section A.05.08.

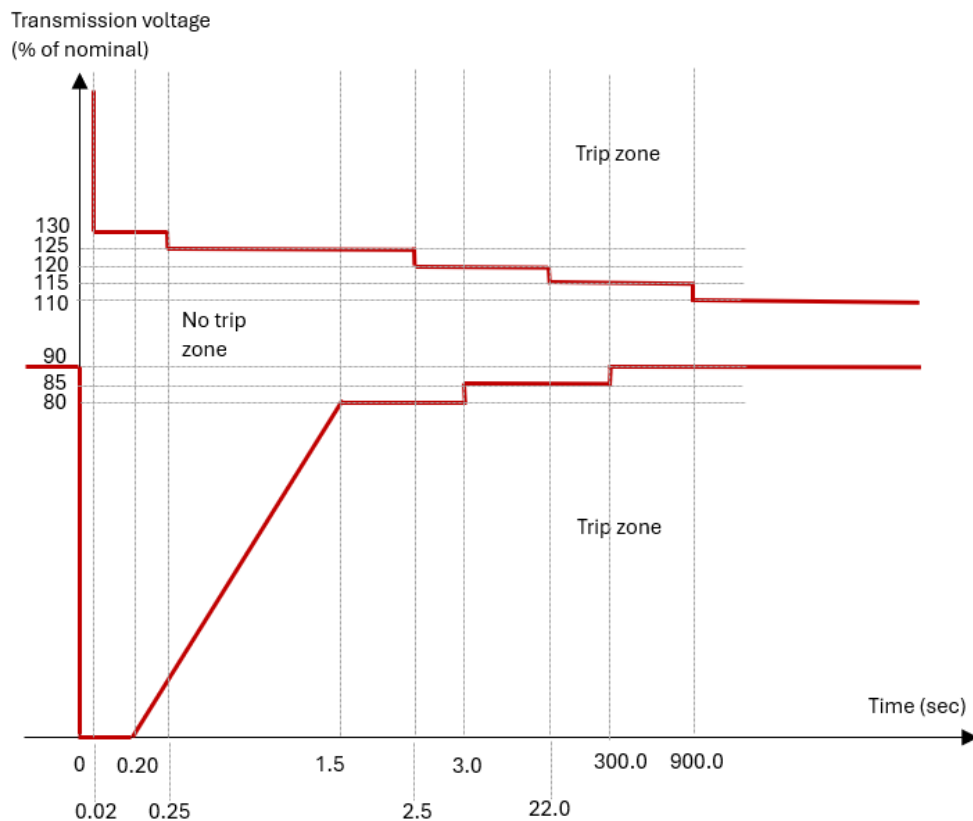
A.05.07. Voltage support capability

- (a) The BESS must be capable of responding on a configurable droop curve (1 – 6%) to either local voltage deviations or at POC, without receiving external control signals from CEB.
- (b) Droop set point will be set based on studies conducted by the CEB as part of the design of the Facility, for the purpose of maintaining network stability.
- (c) The grid-forming energy storage system shall have automatic voltage regulation (AVR) functionality and the system's 5% voltage step response time shall be <50 ms, with overshoot $\leq 30\%$ of the voltage step magnitude.

A.05.08. Voltage withstand and fault ride through capability

- (a) Under voltage variations, the BESS must remain connected and fully operational for duration as shown in the following table and corresponding figure.
- (b) Voltage ride through settings must be configurable in at least ten bands between 0 and 1.3p.u. If default capability of equipment exceeds this requirement, the default capability must be maintained and accessible to CEB without restriction.

Voltage at POC	Time
p.u.	
$v > 1.30$	0.02 seconds
$1.30 \geq v > 1.25$	0.25 seconds
$1.25 \geq v > 1.20$	2.5 seconds
$1.20 \geq v > 1.15$	22 seconds
$1.15 \geq v > 1.10$	15 minutes
0.90 – 1.10	Continuous operation
$0.90 > v \geq 0.85$	5 minutes
$0.85 > v \geq 0.80$	3.0 seconds
$0.80 > v \geq 0.0$	Linear between 0.2 seconds and 1.5 seconds
$v = 0.00$	0.2 seconds



A.05.09. Response to disturbances following contingency events

(a) BESS must remain in continuous uninterrupted operation for any disturbance caused by:

- i. a credible contingency event
- ii. a three-phase fault in a transmission system cleared by all relevant primary protection systems (250 millisecond)
- iii. a two phase to ground, phase to phase or phase to ground fault in a transmission system cleared in:
 - a. the longest time expected to be taken for a relevant breaker fail protection system to clear the fault; or
 - b. if a protection system referred to in subparagraph (a) is not installed, the greater of the time specified by CEB (or if none is specified, 550 milliseconds) and the longest time expected to be taken for all relevant primary protection systems to clear the fault; or
- iv. a three phase, two phase to ground, phase to phase or phase to ground fault in a distribution network cleared in:
 - a. the longest time expected to be taken for the breaker fail protection system to clear the fault; or
 - b. if a protection system specified by CEB, the greater of 1000 milliseconds and the longest time expected to be taken for all relevant primary protection systems to clear the fault,

provided that the event is not one that would disconnect the production unit from the power system by removing network elements from service.

(b) BESS and each of its production units must remain in continuous uninterrupted operation for a series of up to 6 disturbances within any five-minute period caused by any combination of the events described in paragraph (c) where:

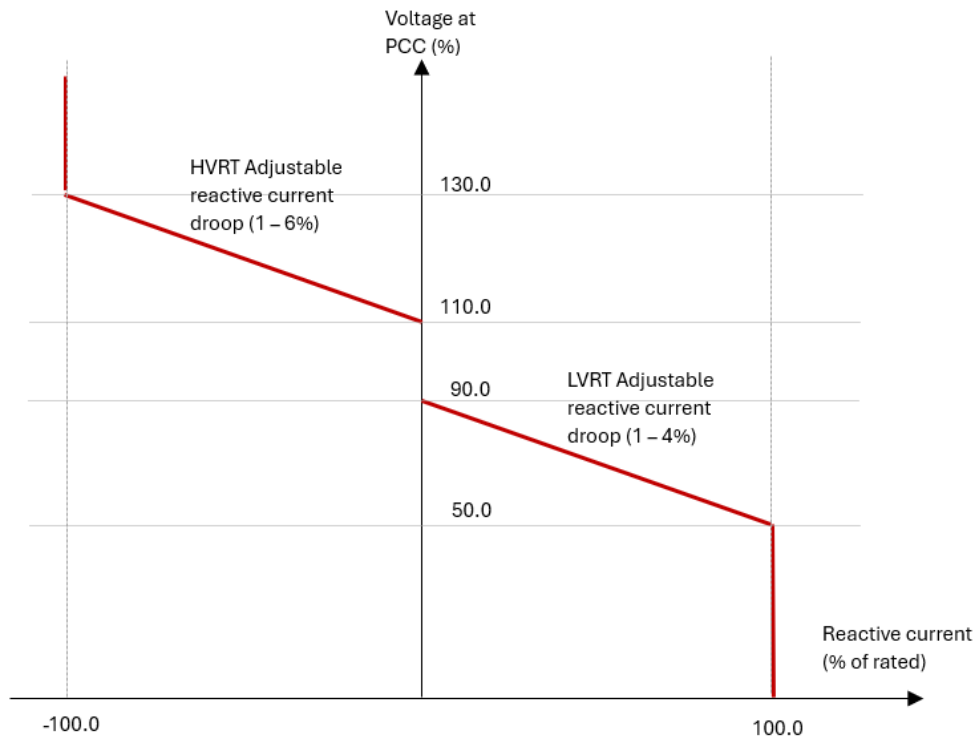
- i. up to four of the disturbances (two LLLG faults, one LL and one LG fault) cause the voltage at the connection point to drop below 50% of normal voltage
- ii. up to one disturbance is cleared by a breaker fail protection system or similar back-up protection system
- iii. up to one disturbance causes the voltage at the connection point to vary within the ranges under clause A.05.08. .
- iv. provided that none of the events would result in:
 - a. the islanding of the generating system or integrated resource system or cause a material reduction in power transfer capability by removing network elements from service
 - b. the cumulative time that voltage at the connection point is lower than 90% of normal voltage exceeding 1,800 milliseconds within any five-minute period
 - c. the time integral, within any five-minute period, of the difference between 90% of normal voltage and the voltage at the connection point when the voltage at the connection point is lower than 90% of normal voltage exceeding 1 pu second.

A.05.10. Reactive current injection and active power recovery during the fault

(a) Subject to any changed power system conditions or energy source availability beyond the BESS reasonable control, BESS respect of the types of faults described in subparagraphs clause A.05.09. must have BESS capable of supplying to or absorbing reactive current from the network:

- i. to assist the maintenance of power system voltages during the fault:
 - a. BESS inverter shall inject reactive current during the fault (within their capability) based on the amount of remaining system voltage. Amount of reactive current during the contingency shall be equal to 2% for every 1% drop in voltage at the POC as a default value. Further, the BESS unit shall be capable of providing reactive current of x% per 1% drop in voltage at POC where x% shall be set between 1% to 6%.
 - b. Additional reactive current injection during fault shall be in addition to the steady state reactive current given at that time by the BESS.
 - c. during the disturbance and maintained until connection point voltage recovers to between 90% and 110% of normal voltage, or such other range agreed with the CEB; and
- ii. from 100 milliseconds after clearance of the fault, active power shall recover to at least 95% of the level existing just prior to the fault.

- (b) CEB may request the contribution of positive, negative and zero sequence currents depending on the requirements of fault detection near the POC.
- (c) Allowable activation delays are between 20ms - 40ms (in order for ac protection systems to detect faults without undue delays) and provide or absorb the required reactive current within 70 milliseconds



A.05.11. Quality of electricity generated (harmonic, flicker and voltage unbalance)

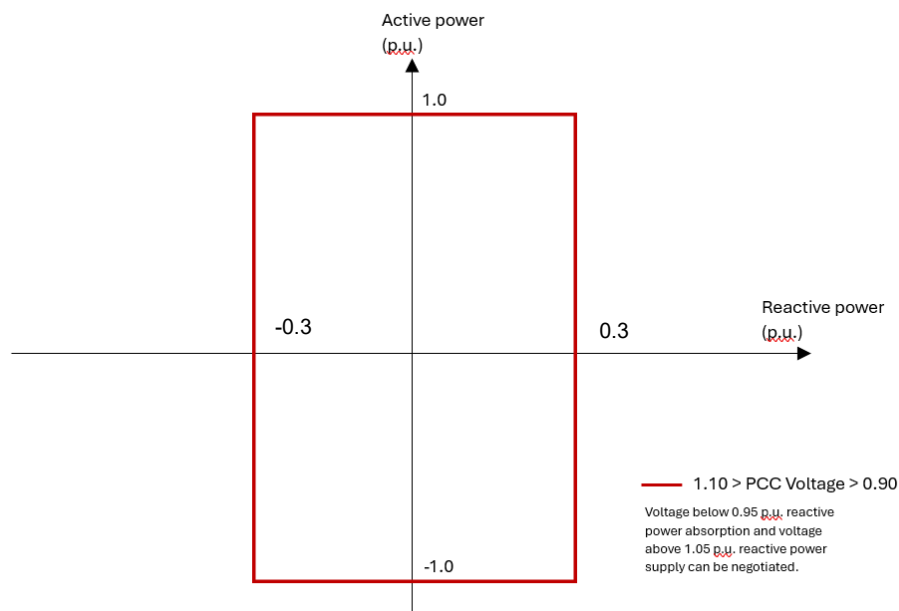
- (a) The BESS shall ensure that when connected to CEB network, it must not produce level of distortion or fluctuation of the supply voltage at the connection point:
 - i. rapid voltage changes greater than 3% of the nominal voltage during the normal operation
 - ii. harmonic voltage distortion greater than the emission limits allocated by the CEB in accordance to 3.17.2.5 of Grid Connection Code.
 - a. harmonic analysis shall consider harmonic order 2 – 50
 - iii. voltage unbalance greater than the limits allocated by the CEB in accordance with IEC 60034-1.
 - iv. voltage flicker greater than the limits allocated by CEB, i.e., $P_{st} = 0.35$ and $P_{lt} = 0.25$.
- (b) The design of the harmonic mitigation measures (including filters if required) shall be demonstrated through a harmonic analysis study.
- (a) Measurement and evaluation of respective emissions shall be carried out in accordance with IEC 61000-3-6. Measurement and evaluation reports shall be prepared in accordance with IEC 61000-3-6 and IEC 61000-3-7.

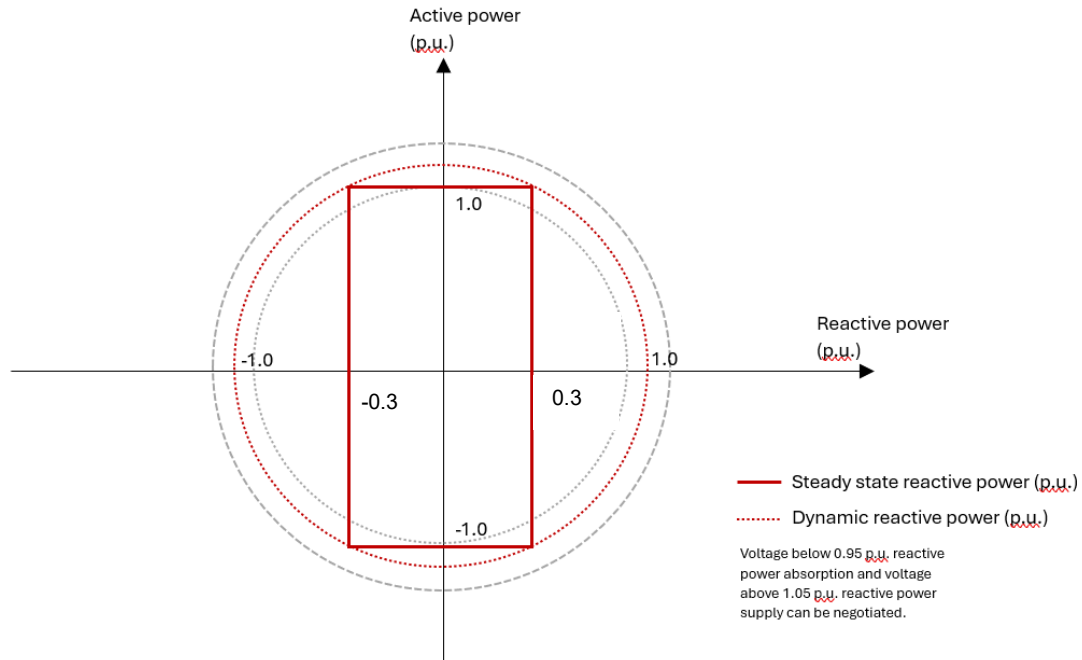
A.05.12. Quality of electricity and continuous uninterrupted operation

- (b) BESS, harmonic filters and associated reactive plants, if any must not disconnect from the power system as a result of voltage fluctuation, harmonic voltage distortion and voltage unbalance conditions at the connection point within the levels specified in clauses (as per the planning limits given in **Error! Reference source not found.**)

A.05.13. Reactive power capability

- (a) The BESS operating at:
- any level of active power; and
 - any voltage at the POC within the limits 0.9 and 1.1 per unit without a contingency event,
- (c) must be capable of supplying and absorbing continuously at its POC (33 kV) an amount of reactive power of at least the amount equal to the product of the rated active power of the BESS and 0.3.
- (d) must be capable to operate in any combination of exporting or importing active and reactive power, otherwise known as four quadrant operation.
- (e) If the BESS cannot meet the reactive power exchange requirements as outlined above, the Project Company shall expect to install additional equipment necessary to meet those requirements. The specific selection and design of such equipment shall be approved by the CEB.





A.05.14. Voltage, reactive power and power factor control modes

- (a) The BESS shall be able to provide automatic control at the POC over the entire operating range of the BESS, without causing system instability. Control modes include:
- Voltage control
 - Voltage droop control
 - Reactive power control
 - Power factor control
- (b) The BESS shall be capable of contributing to voltage or voltage droop control at the POC utilizing its capabilities, while respecting reactive power capability and reactive power exchange with the AC network, in accordance with the following control characteristics:
- Voltage set-point at the connection point shall be capable of being specified within the range (0.9 pu-1.1 pu), either continuously or in steps.
 - It shall be possible to operate voltage control with or without a dead band around the setpoint. The dead band shall be adjustable and agreed with CEB.
 - Voltage droop control mode shall include the capability to change reactive power output based on a combination of a modified voltage set-point and an additional instructed reactive power component (i.e. voltage-reactive power droop). The slope (i.e. voltage droop) shall be adjustable in the range (typically 2% -12%).
- (c) The BESS shall be capable of operating at a designated reactive power value specified by the CEB as long as the same is within the reactive power capability of the BESS.

- (d) The BESS shall be capable of controlling the power factor to a target at the POC, while respecting Reactive Power Capability and Reactive Power Exchange with the network.

A.05.15. Set point response

- (a) The BESS inverter must be able to respond independently to separate command set points from National System Control Centre (NSCC) for the following performance characteristics under normal operating conditions:
 - i. MW
 - ii. Mvar
 - iii. power factor
 - iv. voltage
- (b) Set point response shall be by bias of the relevant droop curve and grid support functionality shall be maintained, within the active and reactive power capability limits of the BESS.
- (c) The behavior of the relevant performance characteristic following receipt of a set point command at the Facility SCADA, subject to any pre-set ramp rate limitations, and limitations on the active and reactive power capability limits of the BESS, shall be as follows:
 - i. Set point response must commence within 50 ms.
 - ii. Commanded set point value must be achieved within 300 ms
 - iii. Overshoot of the commanded set point value shall not exceed 10% of the value of the change in set point value.
 - iv. Settling time, to within 1% of full-scale output for the relevant performance characteristic, must be achieved within 400 ms.
 - v. BESS shall automatically notify NSCC when its state of charge moves outside the normal operating state of charge limits as described in Functional Specifications and reduce the active power set point to zero in a manner pre-agreed with CEB.

A.05.16. Priority to Active or Reactive power contribution

- (a) The BESS shall be capable of operating in the active or reactive power priority modes, as requested by CEB, while meeting continuous operating rating and short-term overload ratings. This includes utilizing the full rating (continuous and short-term overload) of the inverter current capability and operating in zero active power transfer. Taking into account the capabilities of the BESS, CEB may inform on case by case basis whether active power contribution or reactive power contribution has the priority during low or high voltage operation.
- (b) If not specified active power control shall have the priority.

A.05.17. Control interaction with other generating devices

- (a) Control Interaction is a specific issue that can impact the coordinated operation of Power Electronic devices in a local area. Control systems of dynamic devices can interact in an undesirable manner resulting in unstable or poorly damped oscillations following system disturbances such as fault recovery.
- (b) The BESS shall design the equipment to avoid such undesirable interactions, and the acceptable operation demonstrated through interconnection studies.
- (c) The BESS shall be capable of contributing to the damping of power oscillations in the connected AC Transmission system. The Project Company shall require to provide all required data control system models including parameters for of any small signal modulation controls such as power oscillation damping (POD) controls, or sub-synchronous oscillation damping controls.
- (d) The grid-forming BESS shall have autonomous damping suppression functionality. When low-frequency oscillations (0.2 Hz–2.5 Hz) occur, the system shall suppress oscillations, with active power variation limited to 10%–30% P_n.
- (e) The BESS shall be capable to inject or absorb active phase jump power into the system immediately due to the change of phase angle in the event of disturbance or failure of the system
- (f) The BESS shall be capable of finding stable operation points with a minimum change in active power flow and voltage level, during and after any planned or unplanned change in the BESS or AC Transmission System to which it is connected.
- (g) The grid-forming BESS shall receive and respond to AGC (Automatic Generation Control) commands from the grid dispatch, with a regulation range of -100% to +100% P_n and steady-state active power deviation $\leq 2\%$ P_n. It shall also receive and respond to AVC (Automatic Voltage Control) commands, with steady-state reactive power deviation $\leq 2\%$ P_n.
- (h) The BESS shall support both grid-following and grid-forming control modes, enabling online switching between the two.
- (i) The grid-following mode shall operate stably under a minimum short-circuit ratio (SCR) of 1.2, while the grid-forming mode shall operate stably under a minimum SCR of 1.0.

A.05.18. Fault current

- (a) the contribution of the BESS to the fault current on the connecting network through its connection point must not exceed the contribution level that will ensure that the total fault current can be safely interrupted by the circuit breakers of the connecting network and safely carried by the connecting network for the duration of the applicable breaker fail protection system fault clearance times, as specified for the relevant connection point by the CEB
- (b) BESS connected plant must be capable of withstanding fault current through the connection point up to the higher of:
 - i. the level specified by CEB as per the clause A.04.
 - ii. the highest level of current at the connection point that can be safely interrupted by the circuit breakers of the connecting network and safely carried by the connecting network for the duration of the applicable breaker fail protection system fault clearance times, as specified by the CEB
 - iii. circuit breaker provided to isolate a generating unit, generating system, bidirectional unit or integrated resource system from the network must be capable of breaking, without damage or restrike, the maximum fault currents that could reasonably be expected to flow through the circuit breaker for any fault in the network or in the generating unit, generating system, bidirectional unit or integrated resource system, as specified in the connection agreement.

A.05.19. Loss of communications

- (a) In the event of loss of communications, the set point response of the BESS shall automatically revert to a state as agreed with CEB during design of the Facility.
- (b) Upon restoration of communications, the BESS plant shall not automatically restore operations but shall assess and issue ready status. Restart of operations shall only be commenced following instruction from CEB.

A.05.20. Safe shutdown

- (a) In the event network fault that leads to power system conditions outside the requirements of this specification, the BESS must automatically execute a safe and controlled shutdown.
- (b) Upon restoration of standard operating conditions, the BESS plant shall not automatically restore operations but shall assess and issue ready status. Restart of operations shall only be commenced following instruction from CEB.
- (c) Subject to the limits of the backup power supply of the Facility, communications and essential functions shall remain operational throughout a safe shutdown event.

A.05.21. Protection from power system disturbances

- (a) BESS to be automatically disconnected from the power system in response to abnormal conditions arising from the power system, the relevant protection system or control system must not disconnect the BESS for:
 - i. conditions for which it must remain in continuous uninterrupted operation or
 - ii. conditions it must withstand under the Grid Connection Code; and
 - iii. a relevant system (as defined in paragraph (a)i) connected to a transmission system must have facilities to automatically and rapidly reduce its generation:
- (b) CEB may require the BESS to be automatically disconnected by a local or remote-control scheme whenever the part of the network to which it is connected has been disconnected from the national grid, forming an island that supplies load.
- (c) Notwithstanding clauses in Grid Connection Code or in this schedule, BESS may be automatically disconnected from the power system under any of the following conditions:
 - a. in accordance with an ancillary services agreement between the BESS and CEB
 - b. where a source of load that is not part of the BESS has the same connection point as the generating system and CEB agree that the disconnection would in effect be under-frequency load shedding
 - c. where the generating system or integrated resource system is automatically disconnected under, Protection systems that impact on power system security or by an emergency frequency control scheme
 - d. where the generating system or integrated resource system is automatically disconnected under Protection to trip plant for unstable operation; or
 - e. in accordance with an agreement between the Project Company and CEB (including an agreement in relation to an emergency control scheme) to provide a service that CEB agrees is necessary to maintain or restore power system security in the event of a specified contingency event.

A.05.22. Data and information exchange between BESS and the System Control

- (a) Exchanged Information
 - i. Regarding instrumentation for the operation, the BESS module shall be equipped with an automatic controller capable of receiving instructions from the CEB system control. This automatic controller shall be capable of operating the units of the BESS module in a coordinated way.
 - ii. The following information shall be provided to CEB by the Project Company.
 - a. Typical SCADA signal requirement as per the System Control.

- b. Detailed high resolution forecast data to be sent to System Control for the PPM having curtailment facility.
 - iii. The BESS controller shall be able to receive the following signals and commands from CEB based on communication with CEB. CEB shall specify the variable type and quality requirements of the supplied signals.
 - a. Start-up command
 - b. Active power order set-point
 - c. Frequency sensitive mode settings
 - d. Reactive power, voltage, or similar set-points
 - e. Reactive power control mode
 - f. Power oscillation damping (POD) control settings, if applicable
 - g. Synthetic inertia, if applicable
 - h. Normal and emergency blocking command
 - i. Ramp blocking command
 - j. Remedial Action Schemes status and settings
 - k. Active power flow direction command, if applicable
 - l. Fast active power reversal command, if applicable
 - m. Redundancy reductions and auxiliary power single source status
 - n. Other project-specific set-points and commands, as specified by CEB.
 - iv. The Project Company shall specify any equipment needed to enable the remote selection of control modes and relevant set-points.

A.05.23. Simulation models

- (a) The Project Company shall provide all the relevant information details as listed in “Data and Model Requirement to be Complied by Inverter based RE Plants prior to be Interconnected to Transmission Licensees Network”(Annex I).
- (b) The Project Company shall provide models that accurately represent the dynamic response of the BESS. The model shall include all site-specific protection, control and other parameter settings as applicable. Both RMS-type and EMT-type models shall be in PSS®E and PSCAD™/EMTDC™ formats, respectively. Further, the Project Company shall supply RMS and EMT PowerFactory models to CEB. The proprietary information of the BESS that forms the model may be protected. However, the models shall have the flexibility to change parameters and select options that CEB will have access to with the field equipment.
- (c) The PSS®E model shall be capable of running in version 35.x while PSCAD™ model shall be capable of running in Version 5.x with Intel Fortran compiler one API version.
- (d) As a minimum requirement, bidders must fulfill one of the following:
 - I. Submit initial non-site-specific simulation models of the BESS in both PSS®E and PSCAD™/EMTDC™ formats along with the tender for evaluation; or

- II. provide test results with the tender, demonstrating the BESS response in terms of voltage (V), active power (P), and reactive power (Q) during deep and shallow faults, and for a +50-degree phase angle step change at the Point of Connection (POC), under Short Circuit Ratio (SCR) levels of 1, 3, 5, and 10, with an X/R ratio of 5.

Failure to meet at least one of the above requirements may result in the rejection of the bidder's technical proposal.

- (e) The Project Company shall provide the study results in accordance with the requirements of the "Dynamic Plant Model Quality and Dynamic Response Test," and demonstrate that the plant complies with all functional requirements specified in this specification. The study results shall be conducted using PSS®E and PSCAD™/EMTDC™, and submitted for review and acceptance by CEB. (Refer Annex A-II). Those simulation studies shall be revised based on the actual system and BESS module tests and adhere to the requirements for the Transmission System and BESS module per the Technical Specifications, as well as the following requirements:
 - i. For the purpose of dynamic simulations, the models provided shall contain at least, but not limited to, the following sub-models, depending on the existence of the mentioned components:
 - a. BESS model
 - b. AC component models
 - c. DC system models
 - d. Voltage and power controller (power plant controller)
 - e. Special control features if applicable (e.g. power oscillation damping (POD) function, sub-synchronous torsional interaction (SSTI) control)
 - f. Multi terminal control, if applicable
 - g. BESS protection models as agreed between CEB and the Project Company
- (f) The Project Company shall verify the models against the results of compliance tests carried out and a report of this verification shall be submitted to CEB within 3 months after the commissioning. The models shall then be used for the purpose of verifying compliance with the requirements of this Regulation including, but not limited to, compliance simulations and used in studies for continuous evaluation in system planning and operation.
- (g) The Project Company shall submit BESS recordings to CEB during commissioning, in order to compare the response of the models with these recordings.

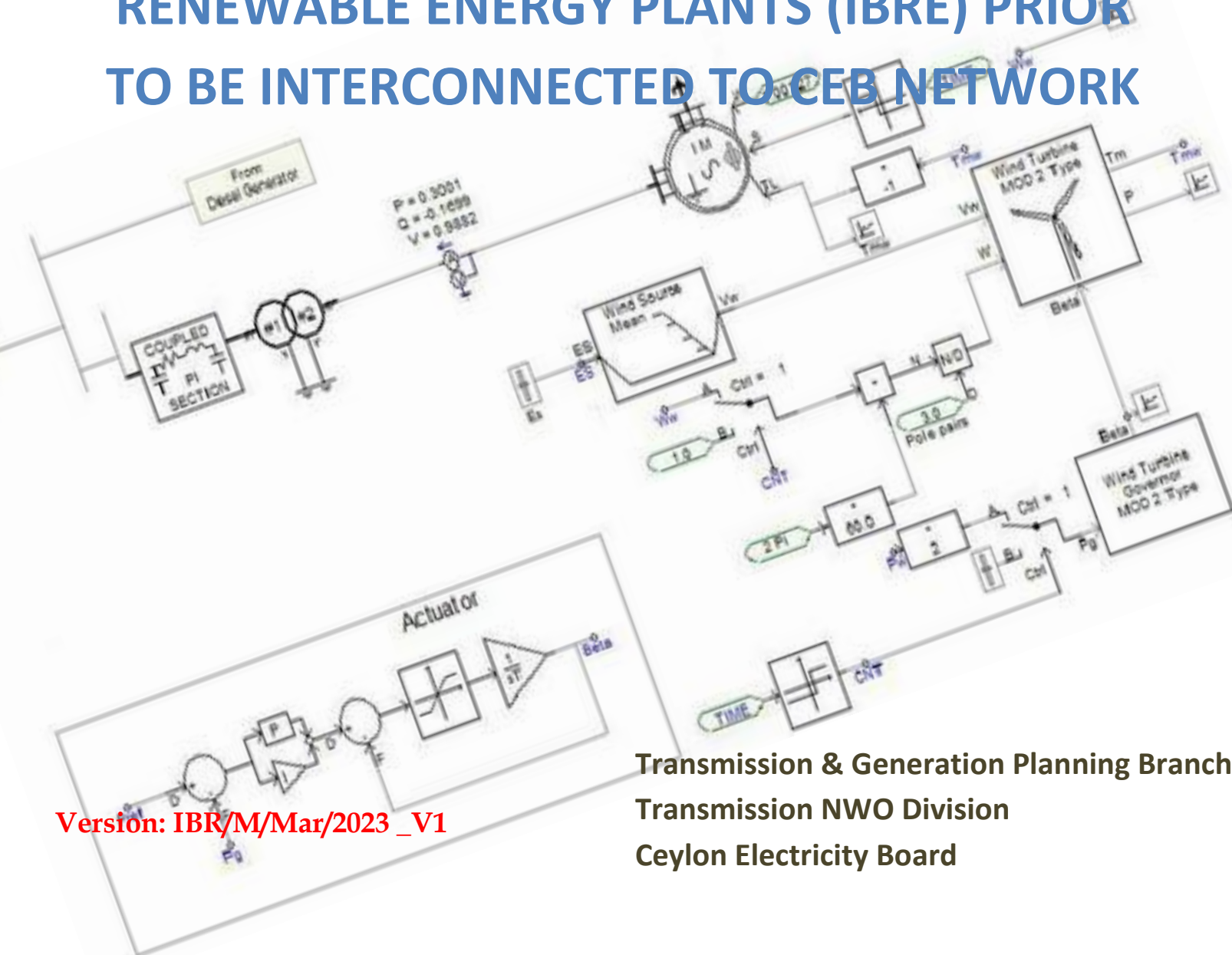
A.05.24. Testing, monitoring and compliance review

The additional requirements considered for BESS

- (a) Testing and compliance
 - i. The BESS shall be designed and tested to meet the requirements of the CEB operational requirements, performance studies and any Interconnection Agreements. Sufficient monitoring shall be provided to CEB to determine that the BESS complies with those performance requirements.
- (b) The Project Company shall provide commissioning test plan two months before testing dates including number of proposed hold points. Commissioning test plan need to be approved by CEB.
- (c) Grid Connection Testing
 - i. In addition to the site acceptance tests, the Project Company shall recommend, provide procedures and carry out compliance tests in line with Best Industry practices. The tests shall include, but not be limited to:
 - a. Harmonic voltage distortion measurements
 - b. Frequency response test
 - c. Reactive power compensation test
 - d. Voltage control test
 - e. Tariff metering tests
 - f. Fault ride through capability tests
 - g. Audible Noise measurements
 - h. Performance Guarantee requirements
 - i. Electro-Magnetic Interference measurements
- (d) These tests shall be performed at different active and reactive power levels (worst case also) to demonstrate the features and the stability of the BESS with respect to CEB's AC system.
- (e) The BESS shall include any additional monitoring and recording required for the purpose of completing grid connection testing in the scope of supply.
- (f) The Project Company shall coordinate grid connection testing with CEB, including notification to all relevant parties and seeking approval for all test procedures prior the commencement of any system tests.



DATA AND MODEL REQUIREMENTS TO BE COMPLIED BY INVERTER BASED RENEWABLE ENERGY PLANTS (IBRE) PRIOR TO BE INTERCONNECTED TO CEB NETWORK



Version: IBR/M/Mar/2023 _V1

Transmission & Generation Planning Branch
Transmission NWO Division
Ceylon Electricity Board

INTRODUCTION

This document contains requirements that must be mandatorily complied by all Inverter based renewable power plants above a given threshold capacity (which would be declared by CEB from time to time) when they submit to CEB the software Model of the proposed power plant (in the format specified in this document) to demonstrate compliance to the requirements of the Grid Code. As inverter based power plants inherently creates additional technical challenges that were not present with synchronous technology based conventional generating plants, it would be to the best interest of both the project developer and CEB to ensure that the proposed power plant could be operated successfully (in compliance with the requirements of the prevalent Grid Code) during varying system conditions that are likely to occur during the operational period of the facility. All project proponents shall ensure that the performance demonstrated by the proposed power plant through this Model are available in the actual plant constructed and commissioned. Once all commissioning has been carried out, the developer shall provide to CEB the accurate As Built Model Software Model (in the format stated here).

1. RMS MODEL REQUIREMENTS (PSSE™)

This Model must be provided in PSSE™ (Power System Simulator for Engineering) format.

The PSSE Model provided by the project proponent of the proposed Power Park Module should be an accurate representation of the overall plant as well as specific components. The Model must contain all sub systems provided by the vendor of each system (vendor specific) and should not use generic representations instead. The Model should be accurate enough to study system level transients where the variation to the frequency range (electromechanical, rms voltage, active and reactive power) can be in the order 0.5 Hz to 5 Hz.

The vendor should provide a Model User's Guide describing various Model details, input parameters and output parameters.

The rms Model should be 'benchmarked against the response of the detailed EMT Model and a Model benchmark report provided to CEB as specified in the CEB document "*Model Quality & Dynamic Response Test*" (MQDRT) document.

The response of the Model should be verified on a PSSE network Model (representative of the CEB network around the point of connection), which would be provided by CEB.

The Model must initialize to the conditions of a corresponding steady state load flow condition.

The RMS Model should run at reasonable time steps (5 ms or higher) used for dynamic stability studies. The Model response should be constant over a range of simulation timesteps.

2. EMT MODEL REQUIREMENTS (PSCAD™)

This Model must be provided in PSCAD™ format.

The PSCAD Model provided should be an accurate representation of the overall plant design as well as specific components. The Model should be vendor specific. The Model should include site specific settings and should not be based on generic settings. The Model should be accurate to study system level transients where the frequency range can be in the order of few Hz to potentially kHz range.

The vendor should provide a Model User's Guide describing various Model details, input parameters and output parameters.

- I. The Model should accurately represent the actual plant design.
 - a. The Model should include plant level controls as well as the inverter level (inner loops) controls loops that derive the firing pulses for valve switches.
 - b. The input signal measuring and filtering modules should be based on actual signal processing algorithms and filters as in the physical implementation. Signal processing and communication delays must be represented in the Model.
 - c. The power electronic switches may be represented as switching type Models, detailed equivalent method or through controlled voltage or current source (average type) representation. If the Model is based on average type representation of the inverters, the

proponent should verify that the control and protection functionalities are not simplified, and the Model is suitable for dynamic response analysis where the transients can be in the frequency range identified above.

- II. The Model should include all plant level and inverter level control and protection functions as implemented in the actual equipment.
 - a. Voltage and frequency protection settings,
 - b. Fault ride-through activation and deactivation settings
 - c. Active and reactive current injection/absorption settings during a fault
- III. The Phase locked loop (PLL), the DC link chopper and DC link protection and inverter current limiting functions (and other FRT related settings) are known critical designs that impact the fault ride through (FRT) response, overall plant stability and grid code compliance.
- IV. The Model should include both electrical and mechanical (turbine characteristics in the case of wind turbine generator Models) component details as applicable.
- V. The Model should provide access to all inputs and parameters that are accessible to the end user.
 - a. Selection of operating mode (example: voltage control, Q control of power factor control)
 - b. Access to user adjustable settings such as plant power and voltage references, droops, ramp rates, fast frequency response mode settings and dead bands
 - c. Ability to enable or disable specific control options (example: damping controls, fault current injection)
- VI. Model should provide access to internal signals that are required by system study engineers to interpret Model response and complete system level studies.
 - a. Trip and alarm signals with appropriate description in Model User's Guide
- VII. The vendor should specify if the EMT Model provided is suitable for harmonic impact studies. The vendor will provide alternate Models for harmonic studies if applicable.
- VIII. PSCAD simulation specific details
 - a. The Model should be compatible across a range of FORTRAN compilers and PSCAD software versions as agreed with the end client.
 - b. The user will have the flexibility to change the simulation time step. The simulation time step should not be hard-coded. It is expected that the Model will be able to produce accurate dynamic response results at a time step of 10 us and higher. The vendor must specify the recommended range of simulation time step.
 - c. The Model should be 'scalable' to represent a number of units operating in parallel.
 - d. It is expected that the Model can reach the expected (user setting) steady state in less than 3s from the start of the simulation. The vendor must make necessary efforts to ensure that the Model will be compatible with the 'snap shot' feature of PSCAD.
 - e. The user should be able to use multiple instances of the Model in a single simulation setting (without having to revert to parallel core simulations)
- IX. Other details

- a. Clearly identify the manufacturer's EMT Model release version and the applicable corresponding hardware firmware version.
- b. Provide an illustrative simulation example where the plant Model is connected to an equivalent Thevenin voltage source. The Model should display stable fault recovery when the Thevenin' source impedance is set to represent the minimum short circuit current condition at the point of connection (POC) and show acceptable response for tests outlined in CEB -MQDRT document.

3. HARMONIC PERFORMANCE MODELS AND DATA REQUIREMENTS

The harmonic injection from the plants should be provided by the plant owner in the form of a guaranteed maximum harmonic current injection table covering the range up to the 100th harmonic.

The frequency-dependent impedance to develop a Norton equivalent current source should also be provided for each unit in a power park module or a BESS.

An illustrative harmonic data table is listed below (Table 1).

Table 1 Guaranteed maximum harmonic injection levels.

Harmonic Order	% of rated current	Phase (Deg)
1	100	0
2		
3		
125		

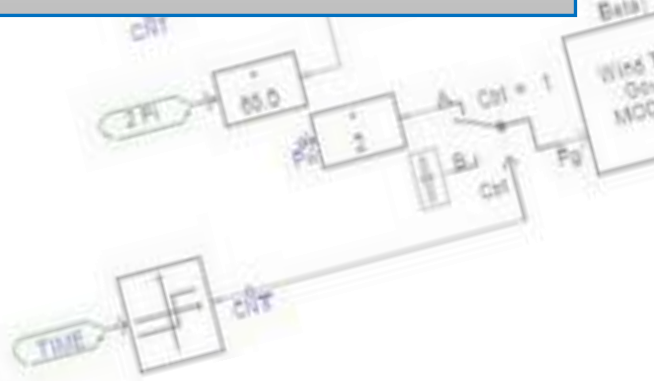
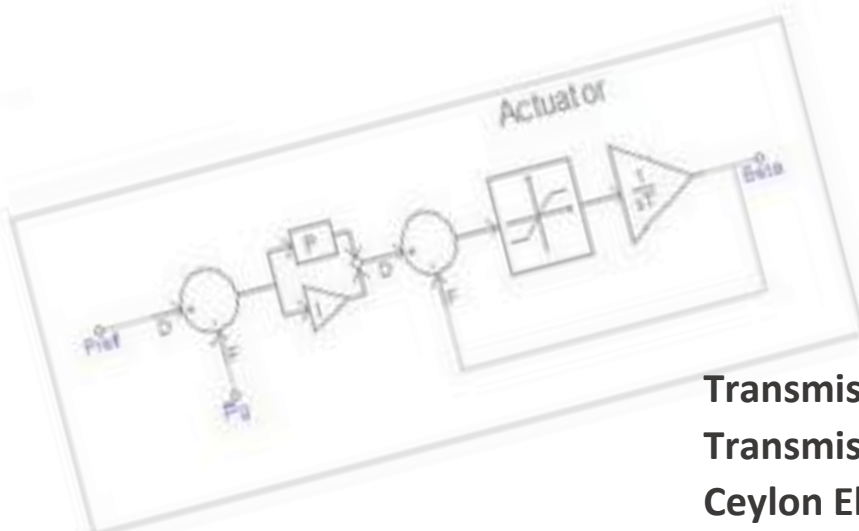
4. POST COMMISSIONING MODEL SUBMISSION

The final response should be tested with field data and also against factory acceptance test data following commissioning. The Model should be adjusted to reflect As Built conditions. The Model should include site specific settings and the Model response should be comparable to field test results.

ලංකා විදුලිබල මණ්ඩලය
இலங்கை மின்சார சபை
CEYLON ELECTRICITY BOARD



DYNAMIC PLANT MODEL QUALITY AND DYNAMIC RESPONSE TEST (MQDRT) TO BE COMPLIED BY INVERTER BASED RENEWABLE ENERGY PLANTS (IBRE) PRIOR TO BE INTERCONNECTED TO CEB NETWORK



Transmission & Generation Planning Branch
Transmission NWO Division
Ceylon Electricity Board

Version: MQDRT /July/2023 _V1 (Ext)

PURPOSE

This Guideline lists the minimum requirements that should be met by a proponent submitting simulation models of plants that are to be connected to the electric network operated by the Ceylon Electricity Board (CEB). The plants include generation, Inverter Based Resources (IBR) or other major equipment for which the connecting party is expected to provide simulation models in PSSE and PSCAD format. The requirements listed in the document are in addition to the general model feature requirements listed in the 'Model Requirements' document published by the CEB.

The document outlines the model response tests that should be performed, test procedure to be followed and the test results that should be submitted along with the software models for CEB review and approval.

PURPOSE	1
1.0 Purpose and scope	1
2.0 Related policies and procedures	2
3.0 Model Acceptance Test Criteria	2
4.0 Model Acceptance Tests	4
4.1 Robustness of the model	4
4.2 Steady State Response	5
4.3 Dynamic Response:	6
4.4 Robustness of the control system and the overall plant design	8
4.5 Response under large disturbances	8
4.6 Specific tests to verify compliance requirements	10
5.0 Model Benchmark Tests -PSS®E and PSCADTM/EMTDC™ Model Benchmarking	11
6.0 Documentation of test result	12

1.0 Purpose and scope

CEB is expecting a rapid expansion of the Sri Lankan Power system with high penetration levels of Inverter Based resources (IBR). Rapid expansions and the expected addition of Wind and Solar PV generation will present significant technical challenges that should be identified at planning and operational stages. To this effect, it is critical that CEB maintains an accurate network model of the Sri Lankan power system. In an IBR heavy power system, the dynamic response characteristics of IBR plants will have a significant impact on overall stability and security of the power system. In addition, CEB will have to ensure that the specific plant will operate stably, meeting dynamic response characteristics outlined by CEB in the Grid code and other applicable documents. Thus, CEN will be depending on the third-party proponents who are connecting plant and equipment to the CEB network to provide accurate and robust simulation models that close captures the repose of the plant under a variety of potential operating conditions.

The main objective of the tests outlined in the document is to ensure that the models provided are,

- Robust and compliant with CEB model requirements (as outlined in Power System Model Guidelines)
- Accurate representation of the plant response under a range of reasonable operating conditions
- The plant design meets the minimum dynamic response requirements.
- Meet plant characteristics as outlined in the Grid Code and other applicable documents.
- Model documentation and model structure and model data are in accordance with CEB guidelines or established industry practice.
- The models provided will be readily incorporated into the overall network model of the Sri Lankan Power system in order for CEB to perform power system planning, operation and connection studies.

This document covers the Model Acceptance Tests for both Root Mean Square (RMS) and Electro Magnetic Transient (EMT) type models that are to be provided to CEB by the proponent. The models should be provided in PSS®E (RMS) and PSCADTM/EMTDCTM (EMT) software platforms.

The Model Acceptance Tests are performed with the Plant Model connected to a simplified representation of the rest of the system. Thus, it is important to note that the submission and subsequent acceptance of the MAT report by CEB does not imply that,

- The plant design meets final compliance and acceptance.

- The Models provided are fully compliant – The models are to be further updated based on design modifications and filed test results during commissioning.

2.0 Related policies and procedures

The models provided should be an accurate representation of the physical plant. In addition to demonstrating the robustness and accuracy of the model through the tests outlined in this document, the model response should be aligned with the Grid code, interconnection contracts as well as protection and measurement reequipments.

Once the models and model test reports are provided to CEB, CEB will inform if the model and the test results are acceptable to CEB.

CEB will not disclose model test results nor specific proprietary details embedded in the model with third parties.

3.0 Model Acceptance Test Criteria

The Model Acceptance Tests are designed to ensure that the models provided are, robust and compliant with CEB model requirements (as outlined in Power System Model Guidelines) and that the model is an accurate representation of the plant response under a range of reasonable operating conditions.

The tests outlined below are designed to verify different aspects of the model and the plant response that the model represents.

- Robustness of the model
- Steady state response
- Dynamic response
- Robustness of the control system and the overall plant design
- Response under large disturbances
- Specific tests to verify specific compliance requirements.

The tests that should be performed as minimum for CEB approval are listed in the following sections. All tests will be performed with the plant model connected to a simplified representation of the external CEB network. Unless otherwise directed by the CEB, all tests will be carried out in a Single Machine- Infinite Bus (SMIB) type test setup as shown in **Error! Reference source not found.** and Figure 2

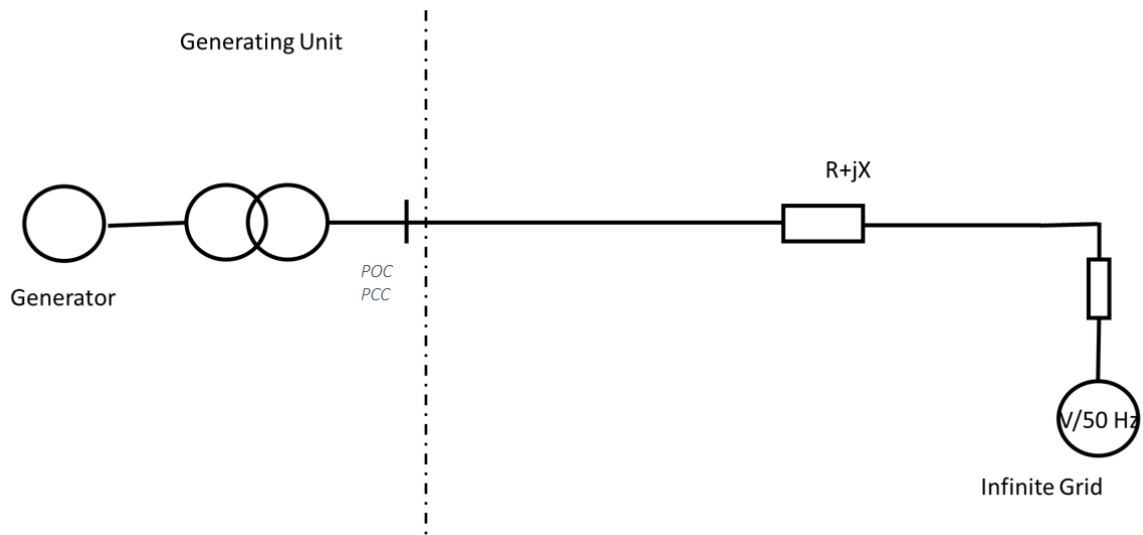


Figure 1 Test setup for plants without Grid substation

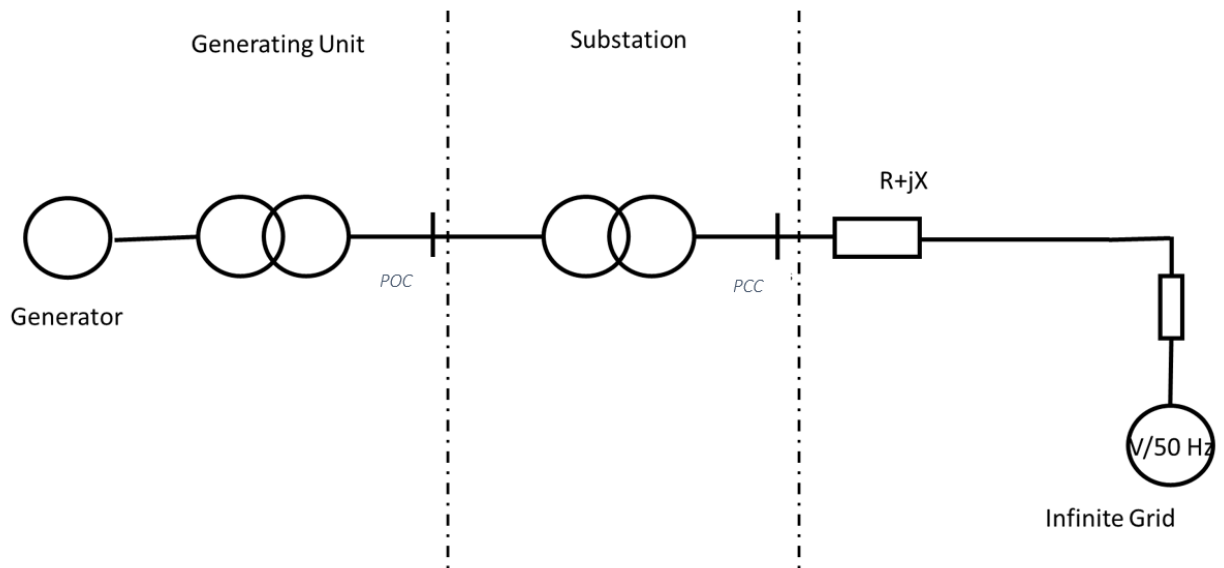


Figure 2 Test Setup for plants with Grid substation (PCC¹— Point of Common Coupling; POC – Point of Connection)

¹ PCC – The project proponent should meet conditions stipulated in the grid code at PCC. These may include Power output, reactive power and voltage conditions.

POC- This is the connection point to the CEB network. PCC and the POC may be the same or different locations.

4.0 Model Acceptance Tests

All tests should be performed on the most current version of the software or as specified by CEB. CEB will also specify the required Fortran compiler compatibility. The POI voltage should be set to 1 PU and the equivalent voltage source frequency set to 50 Hz.

For BESS system, some of the tests may require to be performed under both discharging and charging conditions. Unless specifically stated, the tests for BESS systems are performed under dis-charging condition.

Unless stated specifically, the tests described blow are applicable for both PSS/E and PSCAD.

Note: Input test conditions listed in the tables below should be used unless specific conditions (such as lower SCR levels) are specified by CEB.

4.1 Robustness of the model

Test No	Description of the Tests	Input to the Test	Expected Minimum Recorded Data
T01	The Model should reach the expected study state. • The PSCAD simulation should reach the expected steady state in less than 3 s of simulation time. This test should be performed at 10 us and at the largest time step recommended by the proponent. • The PSSE model should 'flat start' without any uninitialized states	SCR= 3 X/R = 14	Waveform of Voltage (V_{rms}), Active Power (P), Reactive Power(Q) at the PCC
T02	The PSCAD model should be able to start from a saved 'snapshot' file. Run the simulation under T001 with a snapshot taken when the system has reached a steady state.	SCR= 3 X/R = 14	Waveform of Voltage (V_{rms}), Active Power (P), Reactive Power(Q) at the PCC

4.2 Steady State Response

The objective of the steady state tests is to verify that the plant can operate while meeting steady state operating criteria. The steady state tests should be run for a minimum of 5s after the simulation has reached steady state.

Test No	Description of the Tests	Input to the Test	Expected Minimum Recorded Data
T03	With Plant power output at 100%, operate the plant at maximum expected lagging reactive power output. This test may be performed with the plant operating under voltage reference setting or reactive power reference setting	SCR= 3 X/R = 14 P=100% Q= -33%	Waveform of Voltage (V_{rms}), Active Power (P), Reactive Power(Q) at the PCC
T04	With Plant power output at 100%, operate the plant at maximum expected leading reactive power output. This test may be performed with the plant operating under voltage reference setting or reactive power reference setting	SCR= 3 X/R = 14 P=100% Q= +33%	Waveform of Voltage (V_{rms}), Active Power (P), Reactive Power(Q) at the PCC
T05	BESS plants With the BESS charging at 100%, operate the plant at maximum expected lagging reactive power output. This test may be performed with the plant operating under voltage reference setting or reactive power reference setting.	SCR= 3 X/R = 14 P=100% Q= -33%	Waveform of Voltage (V_{rms}), Active Power (P), Reactive Power(Q) at the PCC
T06	BESS plants With the BESS charging at 100%, operate the plant at maximum expected leading reactive power output. This test may be performed with the plant operating under voltage reference setting or reactive power reference setting.	SCR= 3 X/R = 14 P=100% Q= +33%	Waveform of Voltage (V_{rms}), Active Power (P), Reactive Power(Q) at the PCC
T07	Continuous operation under over voltage conditions. This test should be performed with the power output at 100%. The POI voltage should be set to 1.1 PU. The plant may be operated in either Q or V reference control mode.	SCR= 3 X/R = 14 P=100% $V_{pcc} = 1.1pu$	Waveform of Voltage (V_{rms}), Active Power (P), Reactive Power(Q) at the PCC
T08	Continuous operation under low voltage conditions. This test should be performed with the power output at 100%. The POI voltage should be set to 0.9 PU. The plant may be operated in either Q or V reference control mode.	SCR= 3 X/R = 14 P=100% $V_{pcc} = 0.9pu$	Waveform of Voltage (V_{rms}), Active Power (P), Reactive Power(Q) at the PCC

4.3 Dynamic Response:

Test No	Description of the Tests	Input to the Test	Expected Minimum Recorded Data
T09	Voltage step response (High voltage). Once the simulation is in steady state (Power output close to 100%), apply a step change from 1 PU to 1.1 PU to the POI voltage at 5s. This may be achieved by changing the equivalent voltage source magnitude or by changing the transformer tap setting. The plant should be set to voltage control mode. The plant should be capable of operating at full MVA capacity.	SCR= 3 X/R = 14 P=100% Vpcc = 1pu @ 0s Vpcc = 1.1pu @ 5s	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) at the PCC
T10	Voltage step response (low voltage). Once the simulation is in steady state (Power output close to 100%), apply a step change from 1 PU to 0.9 PU to the POI voltage at 5s. This may be achieved by changing the equivalent voltage source magnitude or by changing the transformer tap setting. The plant should be set to voltage control mode. The plant should be capable of operating at full MVA capacity.	SCR= 3 X/R = 14 P=100% Vpcc = 1pu @ 0s Vpcc = 0.9pu @ 5s	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) at the PCC
T11	Voltage step response (High voltage). Repeat T09 with plant in Q control mode	SCR= 3 X/R = 14 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) at the PCC
T12	Voltage step response (Low voltage). Repeat T10 with plant in Q control mode	SCR= 3 X/R = 14 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) at the PCC
T13	PF control mode , Set up the plant to operate in PF control mode with power output at 100%. PF may be set to 1. Change the PF reference to 0.95 lagging and verify the response.	P= 100% PF- 1 to .95 lagging	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) at the PCC
T14	PF control mode , Set up the plant to operate in PF control mode with power output at 100%. PF may be set to 1. Change the PF reference to 0.95 leading and verify the response.	P= 100% PF- 1 to .95 leading	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) at the PCC

Test No	Description of the Tests	Input to the Test	Expected Minimum Recorded Data
T15	Voltage step response (High voltage). Repeat T09 with the external system SCR = 15. The plant should be capable of operating at full MVA capacity.	SCR= 15 X/R = 14 P=100% Vpcc = 1pu @ 0s Vpcc = 1.1pu @ 5s	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) at the PCC
T16	Voltage step response (Low voltage) Repeat T10 with the external system SCR = 15. The plant should be capable of operating at full MVA capacity.	SCR= 15 X/R = 14 P=100% Vpcc = 1pu @ 0s Vpcc = 0.9pu @ 5s	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) at the PCC
T17	Frequency response (over frequency) Once the simulation is in steady state (Power output close to 100%), apply a step change (or a at a rate of change of 4 Hz/s) of 1 Hz to the equivalent voltage source frequency (50 Hz to 51 Hz) at 5s. The power output should remain at pre disturbance level or ramp down if CEB has requested over frequency response to be activated	SCR= 3 X/R = 14 P=100% Fpcc = 50Hz @ 0s Fpcc = 51Hz @ 5s Rate of change of 4 Hz/s	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Frequency (F) at the PCC
T18	Frequency response (under frequency) Once the simulation is in steady state (Power output close to 100%), apply a step change (or a at a rate of change of 4 Hz/s) of 1 Hz to the equivalent voltage source frequency (50 Hz to 49 Hz). The power output should remain at pre disturbance level or ramp up if CEB has requested frequency repose under maximum power output conditions.	SCR= 3 X/R = 14 P=100% Fpcc = 50Hz @ 0s Fpcc = 49Hz @ 5s Rate of change of 4 Hz/s	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Frequency (F) at the PCC
T19	Frequency response (under frequency) repeat T18 with the power output set to less than 100%. The power output should remain at pre disturbance level or ramp up if CEB has requested frequency repose	SCR= 3 X/R = 14 P<100% Fpcc = 50Hz @ 0s Fpcc = 49Hz @ 5s Rate of change of 4 Hz/s	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Frequency (F) at the PCC

4.4 Robustness of the control system and the overall plant design

Test No	Description of the Tests	Input to the Test	Expected Minimum Recorded Data
T20	Phase angle jump Once the simulation is in steady state with plant power at 100%, implement a step change of +/- 50 deg to the POI bus voltage phase at 5s. This can be achieved by changing the phase angle of the equivalent voltage source. The plant should operate stably and continue to maintain pre-disturbance output conditions for P and Q.	SCR= 3 X/R = 14 P=100% Phase angle = 50deg @ 5s	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T21	Step change to power set point. Once the simulation is in steady state with power output close to 100%, change the power reference point to 10%. Once the system has reached the steady state, again change the power reference point close to 100%. The plant output should follow the requested output power. The response time should not be greater than 300 ms. The overall responses (P,Q, Vrms) should be well damped	SCR= 3 X/R = 14 P=~100% 10% ~100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T22	Step change to wind speed or solar irradiation – Once the simulation is in steady state, apply a 25% step change to the wind speed or solar irradiation. If this test cannot be readily implemented due to model limitations, the proponent may obtain CEB consent to omit this test.	SCR= 3 X/R = 14 P=100% Plant operating under a Power set point.	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC

4.5 Response under large disturbances

The following tests should be performed at SCR =3 (unless CEB identifies a lower SCR based on the POI short circuit conditions) and an X/r = 10. The plant should recover from the fault and reach pre fault operating condition in less than 300 ms.

Test No	Description of the Tests	Input to the Test	Expected Minimum Recorded Data
T23	Balanced 3-ph to ground fault applied for 0.12 ms at the POI.	SCR= 3 X/R = 10 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q)

Test No	Description of the Tests	Input to the Test	Expected Minimum Recorded Data
			and Phase angle at the PCC
T24	Balanced 3-ph to fault applied for 0.15 ms at the POI. The remaining voltage at POI = 20%. The fault impedance should be reactive.	SCR= 3 X/R = 10 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T25	Balanced 3-ph to ground fault applied for 0.12 ms at the POI. The remaining voltage at POI = 60%. The fault impedance should be reactive.	SCR= 3 X/R = 10 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T26	Phase AB- ground fault applied for 0.12 ms at the POI.	SCR= 3 X/R = 10 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T27	Phase A- ground fault applied for 0.12 ms at the POI.	SCR= 3 X/R = 10 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T28	Balanced 3-ph to ground fault applied for 0.12 ms at the POI.	SCR= 15 X/R = 10 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T29	Phase AB- ground fault applied for 0.12 ms at the POI.	SCR= 15 X/R = 10 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T30	Phase A- ground fault applied for 0.12 ms at the POI.	SCR= 15 X/R = 10 P=100%	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC

4.6 Specific tests to verify compliance requirements.

CEB may identify specific tests to verify that the plant design complied with CEB requirements. Some specific tests are listed below.

Test No	Description of the Tests	Input to the Test	Expected Minimum Recorded Data
T31	Verification of current injection during low voltage conditions – The proponent should verify that the plant injects reactive current as outlined by CEB in the connection contracts/ PPA. This can be realized by plotting the instantaneous current waveforms under T23 to T30.	Refer T23 toT30	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T32	Fast frequency response – If specified by CEB. The test set is similar to T17, T18 or T19.	Refer T17,T18,T19	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T33	Limited frequency sensitive mode (LFSM_U) – If specified by CEB. The test set is similar to T17, T18 or T19.	Refer T17,T18,T19	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC
T34	Limited frequency sensitive mode (LFSM_O) – If specified by CEB. The test set is similar to T17, T18 or T19.	Refer T17,T18,T19	Waveform of Voltage (Vrms), Active Power (P), Reactive Power(Q) and Phase angle at the PCC

5.0 Model Benchmark Tests -PSS®E and PSCADTM/EMTDC™ Model Benchmarking

CEB will use the PSSE model for a wide range of planning and operational studies. Thus, the PSSE model should be an accurate representation on the plant response for the purpose of such studies. The PSSE model should be benchmarked against the validated PSCAD model.

At a minimum, the response of voltage, power and reactive power will be compared between PSS®E and PSCADTM/EMTDC™ at the inverter terminal and PCC level. The comparison should be acceptable to CEB and ideally lie within the 10% tolerance band. The model should comply with CEB requirements, and any noticeable deviations should be documented to the satisfaction of CEB.

The following DMAT test results of PSS®E and PSCAD/EMTDC are to be benchmarked in the same plot for comparison.

- Voltage reference step test
- Active power reference step test
- Over voltage step response test
- Under voltage step response test
- Balanced fault tests
 - ABC -G fault at POI
 - ABC to ground fault at POI with a remaining voltage.
- Unbalanced fault tests
 - AB-G fault
 - A-G fault

6.0 Documentation of test result

The test results (waveforms should be documented under each applicable test. Where applicable, the proponent should provide comments to explain if the results meet expectations or otherwise.